bulkbuy - MERN Stack - starter models+

Core models

User

- **_id**: ObjectId
- **userId**: string // 16 digits uniqe number we will need a utility for this.
- **firstName**: string
- **lastName**: string
- **role**: enum: customer | administrator | supplier
- **emails**: [{ **address**: string, **verified**: boolean, **primary**: boolean }]
- **phones**: [{ **number**: string, **type**: string, **verified**: boolean }]
- **addresses**: [{ **label**: string, **line1**: string, **line2**: string, **city**: string, **region**: string, **postalCode**: string, **country**: string }]
- **config**: ObjectId (Config)
- **paymentMethods**: [{ **type**: string, **detailsRef**: ObjectId, **last4**: string, **provider**: string, **isDefault**: boolean }]
- **passwardHash**:
- **avatar**: S3file._id
- **status** : active, inactive, deleted, suspended, …
- **metadata**: Map<string, mixed>
- **createdAt**: ISODate
- **updatedAt**: ISODate

Product

- **_id**: ObjectId
- **name**: string
- **descriptions**: [{ **locale**: string, **title**: string, **body**: string }]
- **items**: [{ **itemId**: ObjectId, **salesPrices**: [{ **price**: number, **currency**: string, **from**: ISODate, **to**: ISODate }] }]
- **discountScheme**: object (tiered rules)
- **salesWindow**: { **fromEpoch**: number, **toEpoch**: number }
- **ops_region**: string
- **metadata**: Map<string, mixed>
- **status** : active, inactive, deleted, suspended, on_sale, …
- **reviews**: [ObjectId]
- **createdAt**: ISODate
- **updatedAt**: ISODate

Aggregation

- **_id**: ObjectId
- **itemDtos**: [{ **itemId**: ObjectId, **pricingSnapshot**: object, **supplierId**: ObjectId, **salesWindow**: [{ **from**: ISODate, **to**: ISODate }] }]
- **orders**: [ObjectId]
- **ops_region**: string
- **status** : in_process, pending, processed, suspended,
- **metadata**: Map<string, mixed>
- **createdAt**: ISODate
- **updatedAt**: ISODate

Order

- **_id**: ObjectId
- **userId**: ObjectId
- **items**: [{ **productId**: ObjectId, **itemId**: ObjectId, **pricingSnapshot**: object, **quantity**: number }]
- **orderLocation**: address or geo point
- **deliveryLocation**: address or geo point
- **paymentMethod**: ObjectId or embedded payment reference
- **salesWindow**: { **fromEpoch**: number, **toEpoch**: number }
- **ops_region**: string
- **messages**: [ObjectId]
- **metadata**: Map<string, mixed>
- **status**: enum: draft | submitted | confirmed | cancelled | dispatched | fulfilled
- **createdAt**: ISODate
- **updatedAt**: ISODate

Supply

- **_id**: ObjectId
- **supplierId**: ObjectId
- **requesterId**: ObjectId (optional)
- **items**: [{
  itemId: ObjectId,
  quotes: [{ **pricePerBulkUnit**: number, **numberOfBulkUnits**: number, **discountingScheme**: [{ **minQty**: number, **discountPercent**: number, **description**: string }], **isAccepted**: boolean }]
  }]
- **deliveryLocation**: address
- **status**: enum: quote | accepted | dispatched | cancelled | delivered | received
- **ops_region**: string
- **metadata**: Map<string, mixed>
- **createdAt**: ISODate
- **updatedAt**: ISODate

Message

- **_id**: ObjectId
- **avatar**: S3file._id (optional)
- **type**: enum: issue_wall | email | notification | order | review
- **recipients**: { **all**: boolean, **users**: [ObjectId] }
- **fromUserId**: ObjectId (optional)
- **subject**: string
- **details**: string
- **attachments**: [S3file._id]
- **ops_region**: string
- **status**: enum: draft | submitted | deleted | read | unread
- **replyTo**: ObjectId (optional)
- **metadata**: Map<string, mixed>
- **createdAt**: number (epoch)
- **updatedAt**: number (epoch)

Review

- **_id**: ObjectId
- **reviewerId**: ObjectId
- **revieweeId**: ObjectId
- **productid**: ObjectId
- **itemId:** ObjectId
- **messageId**: ObjectId (Message)
- **rating**: number (1–5)
- **ops_region**: string
- **status** : draft, submitted, deleted, …
- **metadata**: Map<string, mixed>
- **createdAt**: ISODate
- **updatedAt**: ISODate

S3file

- **_id**: ObjectId
- **key**: string (S3 key / object path)
- **metadata**: { **mimeType**: string, **size**: number, **width**: number, **height**: number, **checksum**: string }
- **ops_region**: string
- **status** : in_use, deleted
- **createdAt**: ISODate
- **updatedAt**: ISODate

Config

- **_id**: ObjectId
- **userId**: ObjectId (unique)
- **location**: { **lat**: number, **lng**: number, **address**: string }
- **theme**: enum: light | dark | system
- **isPrivate**: boolean
- **ops_region**: string
- **metadata**: Map<string, mixed>
- **createdAt**: ISODate
- **updatedAt**: ISODate

Item

- **_id**: ObjectId
- **sku**: string (unique, indexed)
- **title**: string
- **slug**: string (unique)
- **description**: string
- **shortDescription**: string
- **brand**: { **id**: ObjectId, **name**: string }
- **categories**: [ObjectId]
- **tags**: [string]
- **images**: [S3file._id]
- **media**: [{ **type**: video|image, **s3**: S3file._id }]
- **price**: { **list**: number, **sale**: number|null, **currency**: string, **effectiveFrom**: ISODate|null, **effectiveTo**: ISODate|null }
- **pricingTiers**: [{ **minQty**: number, **price**: number, **currency**: string }]
- **inventory**: { **stock**: number, **reserved**: number, **backorder**: boolean, **warehouse**: [{ **id**: ObjectId, **qty**: number }] }
- **variants**: [{ **sku**: string, **attributes**: { **color**: string, **size**: string }, **price**: object, **inventory**: object }]
- **weight**: { **value**: number, **unit**: string }
- **dimensions**: { **length**: number, **width**: number, **height**: number, **unit**: string }
- **shipping**: { **class**: string, **freightClass**: string, **shipsFrom**: string }
- **taxClass**: string
- **ratings**: { **avg**: number, **count**: number }
- **reviews**: [ObjectId]
- **relatedProducts**: [ObjectId]
- **seller**: { **id**: ObjectId, **name**: string }
- **metadata**: Map<string, mixed>
- **status** : active, suspended
- **ops_region**: string
- **createdAt**: ISODate
- **updatedAt**: ISODate
- **published**: boolean

Nested models

- **pricing_snapshot**
  - **atInstantPrice**: number
  - **discountedPercentage**: number
  - **discountBracket**: { **initial**: number, **final**: number }

- **email**
  - **address**: string
  - **verified**: boolean
  - **primary**: boolean
  - **verifiedAt**: ISODate

- **phone**
  - **number**: string
  - **type**: string
  - **verified**: boolean
  - **verifiedAt**: ISODate

- **payment_method**
  - **type**: string
  - **provider**: string
  - **tokenRef**: string
  - **last4**: string
  - **expiry**: string

- **description**
  - **subject**: string
  - **text**: string
  - **files**: [S3file._id]

RegionMap

- **_id**: ObjectId
- **ops_region**: string
- **code**: string
- **name**: string
- **description**: { **subject**: string, **text**: string, **files**: [S3file._id] }
- **locations**: [{
    - **locationId**: ObjectId
    - **name**: string
    - **type**: string
    - **description**: { **subject**: string, **text**: string, **files**: [S3file._id] }
    - **address**
        - **line1**: string
        - **line2**: string
        - **city**: string
        - **region**: string
        - **postalCode**: string
        - **country**: string
    - **geo** : [**lat**: number, **lng**: number]
    - **contact:** [**phone**: string, **email**: string]
    - **metadata**: key → value
    - **createdAt**: ISODate
    - **updatedAt**: ISODate
    }]
- **metadata**: key → value
- **createdAt**: ISODate
- **updatedAt**: ISODate

Sales_Window // CRUD methods + methods to add/remove items, update items and item pricing_snapshots. defaults for each item are corresponding values from the last sales_window

- _id
- window[fromEpoch, toEpoch]
- products: {product_id :{items:{item_id, pricing_snapshot,}} }
- metadata
- overflow_id // the id of another sales_window doc to be created if the mongo doc size limit is reached to <= 100kb. create another doc and contiues there. if it full again, create another, sort of linked list chained windows
- createdAt
- updatedAt